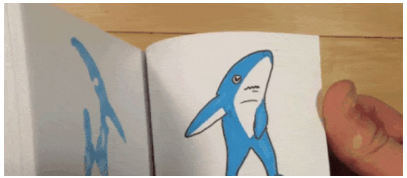
Technical Stuff & Tool Box

What Are We Doing Today? High level



$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Video

series of images

series of matrices

annotated matrices

How to draw an Owl.

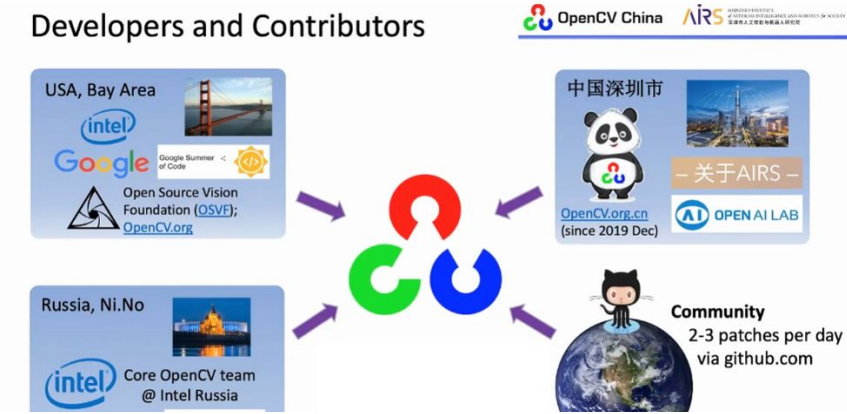
"A fun and creative guide for beginners"

Fig 1. Draw two circles

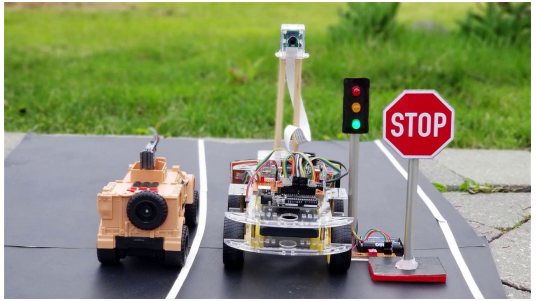
Fig 2. Draw the rest of the damn Owl

OpenCV: What And Why

- Mature (**2000 – present**)
- Free for research and commercial use (**BSD license -> Apache 2 license (5.x)**)
- Implemented in **C/C++** with automatically generated bindings for **Python, Java, Javascript**
- Big, but modular: **>1M** lines of code; **70+** modules (**opencv** + **opencv_contrib**)
- Downloaded **21M** times from SourceForge; **10K** times/week from http://github/opencv



Things Made With OpenCV



Stanley, an autonomous car developed by the Stanford University Racing Team, won the 2005 DARPA Grand Challenge after successfully traversing a 132-mile course. Photo courtesy of the Smithsonian Air & Space Museum.

Self Learning: OpenCV and Beyond

CV Basics

Computer Vision: Crash Course Computer Science #35

Decent 11 minute overview of how the deep learning algos we will automagically run over our images work.

Applied Basics

https://opencv.org/university/free-opencv-course/?utm_source=opcv&utm_medium=menu&utm_campaign=obc <- opencv intro

From the makers of opencv, a series of short how to videos

Mid level data processing

https://numpy.org/doc/stable/user/absolute_beginners.html

With image processing we use numpy instead of pandas, why?

The prime use of dataframes is a matrix like interface for mixed type data, images have one data type, so we don’t need to sacrifice the speed of numpy

Other tools that go with opencv

Numpy

Media pipe

Yolo models

Inspo and Reference

Today’s content

https://github.com/Twin-Cities-Data/Notebooks/tree/main/ImageProcessingKickoff

Computer vision based pose estimation apps

https://learnopencv.com/ai-fitness-trainer-using-mediapipe/

https://learnopencv.com/building-a-body-posture-analysis-system-using-mediapipe/

Video capture docs

https://docs.opencv.org/3.4/dd/d43/tutorial_py_video_display.html

Home work

Main: reproduce this workflow on the provided videos

See if you can get yolo v7 to work instead of mediapipe

Look deeper into media parameters such minimum confidence and see how they impact image annotation

Follow along some of the videos in the opencv bootcamp linked previously

Get ready for the next episode by using the landmarks and connections to calculate angles.